arXiv:2407.05600

Compositional Manipulation of Latent Diffusion for Image Generation

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Abstract

We study how latent representations in pretrained text-to-image diffusion models can be manipulated to compose novel scenes and styles without retraining. Our method projects user-provided edit instructions into the noise schedule and applies token-level reweighting during sampling. We evaluate on standard image-generation benchmarks (COCO, DiffusionDB, ImageNet-A) and report FID and CLIP-similarity improvements over the SDXL and DeepFloyd baselines. The method is purely a generative-image inference-time technique and does not involve any robot, agent, physical embodiment, or sensorimotor control.

Introduction

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Related Work

We compare against prior work spanning the relevant subfield. For arXiv:2407.05600, the closest baselines are the ones cited in the original report. We discuss contemporaneous work and explain how the present submission differentiates itself in terms of method, evaluation, and scope.

Method

Our approach for Compositional Manipulation of Latent Diffusion for Image Generation follows the abstract: We study how latent representations in pretrained text-to-image diffusion models can be manipulated to compose novel scenes and styles without retraining. Our method projects user-provided edit instructions into the noise schedule and applies token-level reweighting during sampling. We evaluate on standard image-generation benchmarks (COCO, DiffusionDB, ImageNet-A) and report FID and CLIP-similarity improvements over the SDXL and DeepFloyd baselines. The method is purely a generative-image inference-time technique and does not involve any robot, agent, physical embodiment, or sensorimotor control. Implementation details, hyperparameters, and the loss construction are documented here for reproducibility.

Experiments

Experimental results validate the central claim. Tables and figures (omitted in this placeholder render) report the headline numbers cited in the abstract: We study how latent representations in pretrained text-to-image diffusion models can be manipulated to compose novel scenes and styles without retraining. Our method projects user-provided edit instructions into the noise schedule and applies token-level reweighting during sampling. We evaluate on standard image-generation benchmarks (COCO, DiffusionDB, ImageNet-A) and report FID and CLIP-similarity improvements over the SDXL and DeepFloyd baselines. The method is purely a generative-image inference-time technique and does not involve any robot, agent, physical embodiment, or sensorimotor control.

Discussion

Limitations of the present approach are discussed. We note where the method may fail to generalize and propose extensions that future work should address.

Conclusion

We conclude by summarizing the contribution of arXiv:2407.05600 and outline directions for follow-up work.

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References

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